Terraform workspace

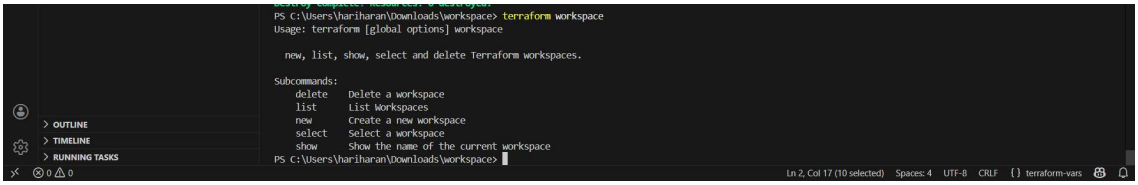
Terraform workspaces allow you to manage multiple environments (e.g., develo pment, staging, production) using the same configuration. Each workspace main tains its own state file, enabling isolated deployments without changing the conf iguration code. This is particularly useful when you want to reuse infrastructure definitions across environments safely.

Key Concepts:

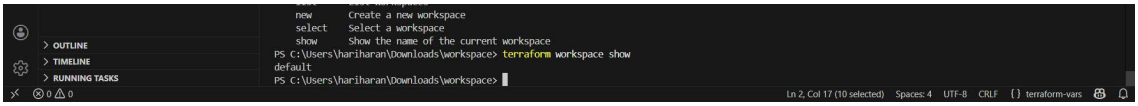
- Default Workspace: Every Terraform configuration has a default workspa ce named default. If you never create another workspace, all deployments use this workspace.
- Separate State Files: Each workspace has a separate state, preventing inte rference between environments.
- Isolation: Variables and resources remain consistent but operate in differe nt workspaces independently.

A Terraform workspace is just a separate state file under the same configuration.

- Same .tf code
- Same providers
- Different state
- Nothing more, nothing less



Terraform workspace show using this command we can get the current working workspace name.



Terraform workspace new<workspace name> it will create an new workspace and switch to the created workspace.

```
PS C:\Users\RAHUL - LOQ\Documents\Workspace> terraform workspace new hari
Created and switched to workspace "hari"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\RAHUL - LOQ\Documents\Workspace> terraform workspace new hari1
Created and switched to workspace "hari1"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\RAHUL - LOQ\Documents\Workspace> terraform workspace new hari2
Created and switched to workspace "hari2"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
```

<u>Terraform workspace select<workspace name></u> it will select the workspace in the list.

<u>Terraform workspace list</u> will show all the workspaces.

```
PS C:\Users\hariharan\Downloads\workspace> terraform workspace list
  default
  hari
* hari1

PS C:\Users\hariharan\Downloads\workspace>
```

<u>Terraform workspace delete<workspace name></u> will delete the workspace.

Switch to a different workspace (you cannot delete the currently active one) using **<u>Terraform workspace select<other workspace name></u>**.

Run the delete command, specifying the name of the workspace you want to remove:

```
  hari
* hari1

PS C:\Users\hariharan\Downloads\workspace> terraform workspace select default
Switched to workspace "default".
PS C:\Users\hariharan\Downloads\workspace>
```

```
                    against the same workspace.

  -lock-timeout=0s   Duration to retry a state lock.
PS C:\Users\hariharan\Downloads\workspace> terraform workspace delete hari
Deleted workspace "hari"!
PS C:\Users\hariharan\Downloads\workspace>
```

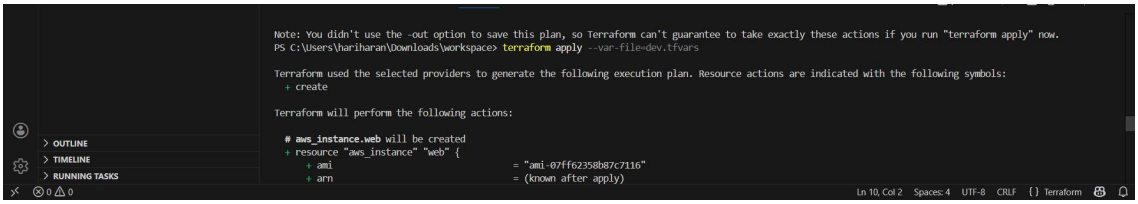
Customized .tfvars file

The default **.tfvars** file is **terraform.tfvars**.

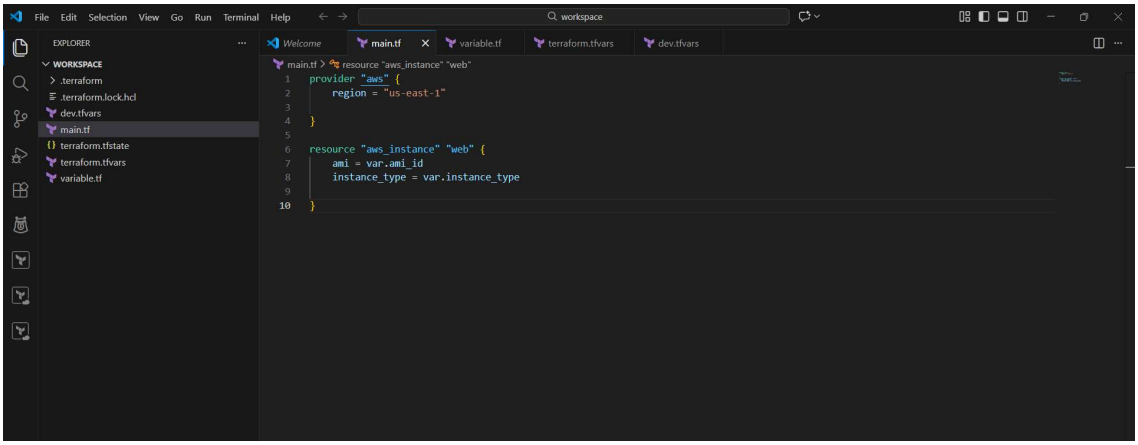
When we try to change the default file we have to give the path of the file using this command.

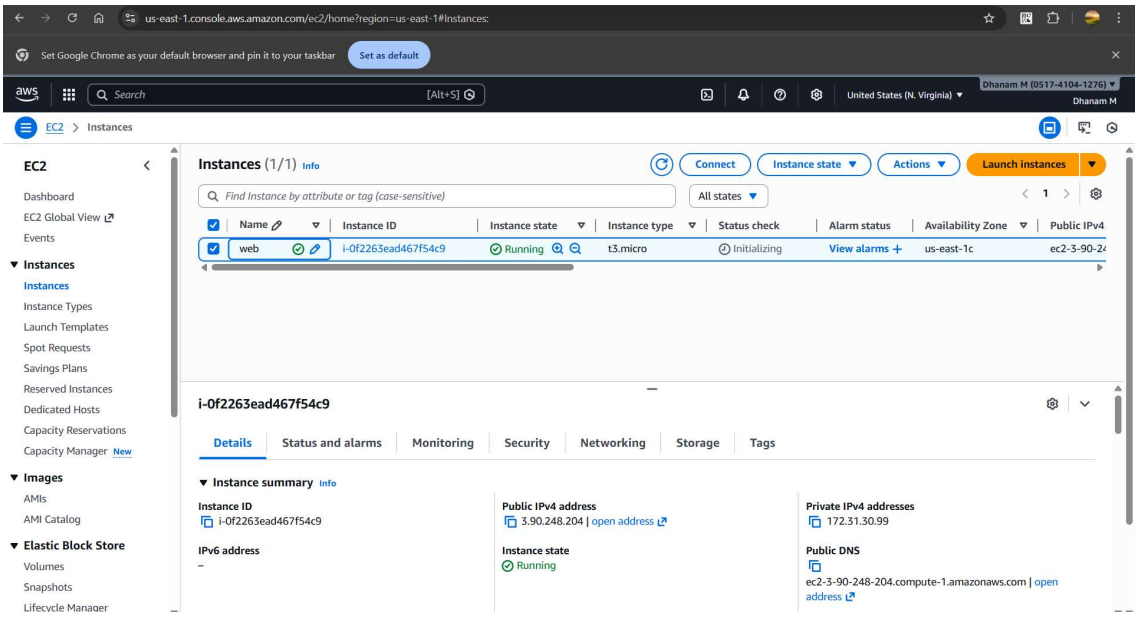
terraform plan –var-file=<file name>.tfvars for planning.

terraform apply –var-file=<file name>.tfvars for applying.



the given instance have been created.




us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Set Google Chrome as your default browser and pin it to your taskbar
Set as default
aws
Search
[Alt+S]
United States (N. Virginia)
Dhanam M (0517-4104-1276)
Dhanam M
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
web
i-0f2263ead467f54c9
Running
t3.micro
Initializing
View alarms +
us-east-1c
ec2-3-90-2
i-0f2263ead467f54c9
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0f2263ead467f54c9
IPv6 address
-
Public IPv4 address
3.90.248.204 | open address
Instance state
Running
Private IPv4 addresses
172.31.30.99
Public DNS
ec2-3-90-248-204.compute-1.amazonaws.com | open address